

# Epsilon-Delta Definition of Continuity

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## 1 Introduction

When we study functions in mathematics, one of the most natural questions we can ask is: *"How does a function behave at a certain point?"*.

Consider a simple example. You are driving a car and you press on the accelerator slightly. Normally we expect the car to accelerate slightly, not teleport to 200 km/h. The output responds proportionally to the input, that is the main idea behind continuity.

A function can fail this expectation in 3 main ways:

- It can **jump** — the output suddenly teleports to a different value.
- It can have a **hole** — the function is undefined at one point.
- It can **blow up** — the output grows without bound near a point.

In each case, a small change in input produces a very big and uncontrollable change in output. Continuity is precisely the condition that rules this out, which means, if a function is continuous at a certain point, it will not cause such behavior.

But saying a function does not break, is not a mathematical statement, it's an intuition. To work with continuity rigorously, to prove it, to build theorems, we need precise definitions. That definition is the  $\varepsilon$ - $\delta$  definition and it's the central object of this document.

Let's try to build it up using words. If a function is continuous at a point  $x_0$ , the closer  $x$  gets to  $x_0$ , the closer  $f(x)$  must get to  $f(x_0)$ . In other words, small changes in input must produce small changes in output. So, the question we can ask is, for any output range we demand around  $f(x_0)$ , can we find an input range around  $x_0$  that guarantees this. If yes, for every possible output range no matter how small, then the function is continuous. This is exactly what the  $\varepsilon$ - $\delta$  definition captures.

## 2 The Epsilon-Delta Definition

Before starting the Epsilon-Delta definition we need to understand one piece of notation. The expression  $|x - x_0|$  means the distance between  $x$  and  $x_0$  on the number line. For example  $|7 - 3| = 4$  and  $|1 - 3| = 2$ , it doesn't matter if the value is negative or on which side  $x_0$  is, the absolute value gives always a positive number. Now let's look at the definition.

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  and let  $x_0 \in \mathbb{R}$ . We say  $f$  is **continuous at**  $x_0$  if:

$$\forall \varepsilon > 0, \quad \exists \delta > 0 \quad \text{s.t.} \quad |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon$$

Translating this word for word means: For all epsilon greater than 0, there exists one delta greater than 0, such that if distance between  $x$  and  $x_0$  is smaller than the delta range, then the distance between  $f(x)$  and  $f(x_0)$  is smaller than the epsilon range.

## 3 Intuition: The Challenger-Defender Game

To understand the definition properly, think of it as a game between two players, a **challenger** and a **defender**.

The challenger picks any output tolerance  $\varepsilon > 0$  and draws a window around  $f(x_0)$ , which is, everything between  $f(x_0) - \varepsilon$  and  $f(x_0) + \varepsilon$ . They are simply saying that every output near  $x_0$  must land inside this window after being transformed by the function.

The defender must respond by finding an input tolerance  $\delta > 0$ , drawing a window around  $x_0$ , which is, everything between  $x_0 - \delta$  and  $x_0 + \delta$ . They are claiming that every  $x$  inside their input window will produce an  $f(x)$  inside the challenger's output window.

The defender wins if their claim is true, while challenger wins if they can find even one  $x$  inside the input window whose  $f(x)$  escapes the output window.

The function is **continuous** at  $x_0$  if the defender can always win, for every possible  $\varepsilon$  the challenger picks. The function is **discontinuous** at  $x_0$  if the challenger can find a small enough  $\varepsilon$  that no  $\delta$  works, which happens exactly when the function jumps or breaks at that point.

## 4 Proving Continuity

Before writing a proof, we always do **scratchwork** first. The goal of the scratchwork is to find  $\delta$  in terms of  $\varepsilon$ . We do this by starting from the output condition and working backwards until we see what condition it requires on the input.

Once we have  $\delta$ , we write the formal proof in the opposite direction — starting from the input condition and arriving at the output condition.

The function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = 2x + 1$  is continuous at  $x_0 = 3$ .

## 4.1 Scratchwork

We have  $f(3) = 2(3) + 1 = 7$ , so the point is  $(3, 7)$ ,  $x_0$  being 3 and  $f(x_0)$  being 7. We want to find  $\delta$  such that whenever  $|x - 3| < \delta$ , we get  $|f(x) - 7| < \varepsilon$ , so we start from the output condition:

$$\begin{array}{ll}
 |f(x) - 7| < \varepsilon & \\
 |(2x + 1) - 7| < \varepsilon & \text{substitute } f(x) = 2x + 1 \\
 |2x - 6| < \varepsilon & \text{simplify} \\
 2|x - 3| < \varepsilon & \text{factor out 2} \\
 |x - 3| < \frac{\varepsilon}{2} & \text{divide both sides by 2}
 \end{array}$$

So the output condition is satisfied exactly when  $|x - 3| < \varepsilon/2$ . We therefore choose  $\delta = \varepsilon/2$ .

## 4.2 Proof

Let  $\varepsilon > 0$  be given. Choose  $\delta = \varepsilon/2$ . Suppose  $|x - 3| < \delta$ . Then:

$$|f(x) - 7| = |(2x + 1) - 7| = |2x - 6| = 2|x - 3| < 2\delta = 2 \cdot \frac{\varepsilon}{2} = \varepsilon$$

Since  $\varepsilon > 0$  was arbitrary,  $f$  is continuous at  $x_0 = 3$ .

Notice the structure. The scratchwork went backwards, from the output condition to the input condition, to discover  $\delta$ . The proof went forwards, from the input condition to the output condition, to verify that  $\delta$  works. The algebra is the same in both directions. The scratchwork is how you think, the proof is how you present.

## 5 Conclusion

In this document we built the  $\varepsilon$ - $\delta$  definition of continuity from the ground up, starting from the simple intuition that a continuous function has no breaks or jumps and turning that intuition into a mathematical condition. We saw it in action through one proofs that succeeded, constructing a valid  $\delta$  for every  $\varepsilon$ . Mastering this definition is not just an exercise, it is the foundation on which all of real analysis is built.